Finding solar panels in 15-minute grid consumption: a model built on physics

Team SilCrst · AIL × Neural Wave challenge, October 2026

1 Problem and approach

For each customer we receive one series: the energy taken from the grid every 15 minutes, which we call the *withdrawal*. From it we must answer four questions. Does the customer have a photovoltaic (PV) system? When was it installed? How large is it, in kWp? Was it extended later? The last question is a bonus.

The score is $F_1 \times (70 + 15D + 15P)$. Detection ($F_1$) counts most. The date score D and the capacity score P only count on the systems we detect. Capacity was the hardest part.

Only 40 customers are labelled, and 22 of them have PV. That is too few to train and check a black-box model. We built instead a simple model around one physical fact (Section 3.1), in plain Python (NumPy, pandas, SciPy, scikit-learn, CatBoost); only the initial capacity comes from a neural network trained without labels (Section 3.4). The 1,149 unlabelled series served to create training examples and to tune thresholds. The final model scores 89.39 in leave-one-out (F_1 1.000, D 0.732, P 0.561).

2 Data and validation

We store each series as a table of days × 96 quarter-hours in UTC. In UTC the sun does not move when the clocks change, so daylight saving time needs no special handling. IDs are read as exact text (reading `NA` as missing would cost about 4 points). The model uses no weather, irradiance or location data, and the delivered code contains no LLM.

We check every change in three ways:

- **Leave-one-out** on the 40 labelled customers: each one is predicted by a model fitted on the other 39 (main threshold included), and scored with the official metric.
- **Simulated PV systems**, where the true answer is known exactly (Section 3.4).
- **Statistics on the unlabelled customers.** We have no labels for them, but they show how a rule behaves at scale. For the second detection rule, for example, we counted how often it says PV among the 844 customers in which the main rule finds none (Section 3.2).

The last check matters because 18 customers without PV are too few to set a threshold. For the second rule and the safety checks, we set each threshold by its cost: the number of real or simulated PV systems it would wrongly remove.

3 The model

3.1 The key idea

At night a building uses a minimum amount of energy: its *standby*. Without PV, the withdrawal never goes below this level. With PV, the panels cover part of the load on sunny days, and the midday withdrawal drops below the standby, often to zero. We call a day *PV-like* when at least 30% of the quarter-hours between 09:00 and 14:00 UTC are below half of that night's standby. Figure 1 shows the pattern: after the installation, blue hours appear around midday, wider in summer and narrower in winter.

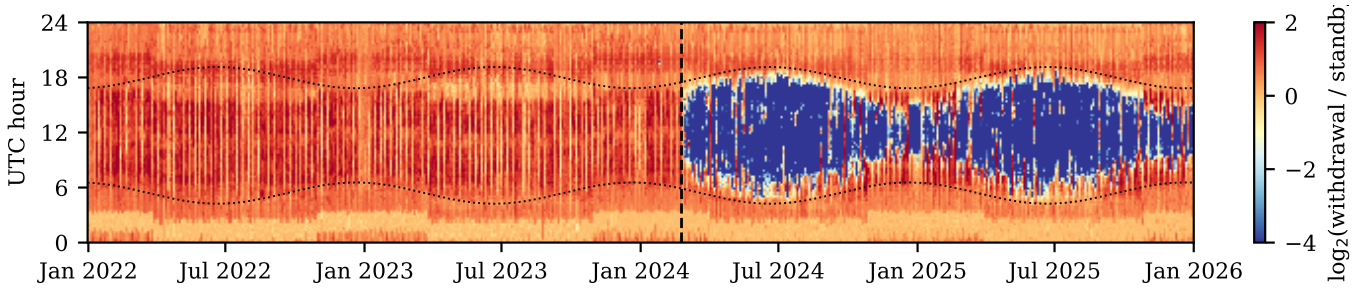


Figure 1: One labelled customer by day (x) and hour (y). Colour: withdrawal divided by the night standby (blue: below standby). Dashed: labelled installation date. Dotted: hours in which panels can produce, estimated from the data.

3.2 Detection

Main rule. A customer has PV if, in some 60-day window, at least 12.5% of the days are PV-like. This threshold is fitted on the labels: it sits in the middle of the gap between the labelled customers without PV (at most 10%) and those with PV (at least 15%, apart from one small system). A stricter threshold of 25% would detect fewer simulated systems (88.7% instead of 92.3%) and miss one labelled system.

Second rule, for small systems. One labelled system is too small for the main rule (two in our first version): its withdrawal never goes below standby. The second rule looks for a lasting drop in midday use compared with night use. To accept it, the drop must look like solar production: it follows the shape of a sunny day, and it is the same on weekdays and weekends, since the sun does not follow a work calendar. Our first version, without this check, made

no mistake on the 18 labelled customers without PV, yet said PV for 12–15% of the 844 customers in which the main rule finds none. Real small systems or false alarms? At the same threshold, the shape check keeps about 2 in 5 of the small simulated systems the rule finds, but only 1 in 30 of the flagged customers: almost all flags were false alarms. The current rule flags 1.1% of the 844 and still finds the small labelled system. On the labelled customers each rule separates PV from non-PV with an AUC of 0.97 (1 is perfect, 0.5 chance); together they find all 22 systems.

Safety checks. Three checks remove detections that cannot be PV (the last two act on the main rule only):

- *Dead meter.* The meter reads exactly zero in at least 40% of night quarter-hours. A real PV customer always has a standby. Cost: 0.7% of simulated systems.
- *Weather.* Customers in the same area share the same sunny and cloudy days, so their PV drops move together. A detection is rejected when neither its midday use nor its below-standby hours follow this shared weather, unless it is clearly seasonal. Cost: 2 of 2,524 simulated systems, and no labelled one.
- *Clock time.* The sun keeps UTC time; people keep local time. A drop that moves by one hour when the clocks change, or sits outside 08:30–15:00 UTC, and that also ignores the weather, is a work schedule, not PV. Cost: none of 898 simulated systems and no labelled one.

The labelled customers contain none of these cases (the dead-meter statistic has an AUC of 0.54 on them), so the checks do not change the leave-one-out score. They act on the unlabelled data: the weather check alone removes 30 of 301 detections.

3.3 Installation date

We place the date where the PV-like days begin, allowing for fewer of them in winter. This fails when a system starts weakly. A high summer load or a weak winter sun can keep only 5–25% of the midday quarter-hours below standby for months, and the date then comes out 1–2 months late. Two fixes address this:

- look back up to 150 days for the start of a steady run of weak PV days (at least 8% of the midday below standby); this can only move the date earlier;
- if the second rule's drop passes its shape check and starts more than 30 days earlier, use its date.

Together they raise D from 0.685 to 0.732 in leave-one-out, and from 0.796 to 0.867 on simulated systems. The median error is now 1 day, and 16 of 22 dates are within a week. Two of the three errors above 30 days are official dates set weeks after the panels started producing, a delay that nothing in the data predicts.

3.4 Capacity

The difficulty. Once PV covers demand, grid withdrawal stays at zero even with larger panels. This *saturation* hides capacity, so we must compare patterns across many days.

Training examples. We supplement the 22 labelled systems by subtracting simulated PV production from real series classified as PV-free. After a chosen installation day:

$$x'_{d,t} = \max\bigl(0,\ x_{d,t} - C \cdot 0.75 \cdot 0.25 \cdot s_d(t) \cdot \sigma_d\bigr).$$

Here x and x' are 15-minute withdrawals in kWh, C is capacity in kWp, 0.75 is the nominal peak-power factor, and 0.25 converts power to quarter-hour energy. The clear-day shape $s_d(t)$ and sunshine factor σ_d come from consumption data. We fit the shape on 27 small PV systems whose output is consumed on site, where the withdrawal reduction reveals production. Clipping at zero reproduces saturation.

CatBoost baseline. Five tree models, each trained on labelled customers and 1,360 simulated systems, estimate capacity from features such as nighttime load and midday-reduction duration. They reach capacity score $P = 0.486$ in leave-one-out, versus 0.31 for a linear model using only real labels.

Final estimator. A vision transformer summarizes each day's 96 readings and compares consumption before and after installation, while retaining the building's load level. It first learns to reconstruct hidden days in unlabelled series, then to predict capacities simulated in 369 PV-free customers. Neither stage uses the 40 development customers' readings or labels. It selects capacity to maximize its expected official score and runs in NumPy. Our rules still determine detection and dates. This raises P to 0.561, but resampling gives a gain interval of -0.06 to $+0.22$: improvement over CatBoost remains uncertain, and simulated examples do not fully represent real systems.

Expansion exception. For small systems revealed by a later expansion (Section 3.5), we keep the rule's initial-capacity estimate and CatBoost's post-expansion estimate. On 147 simulated cases, the transformer underestimated the expanded size ($P = 0.30$, versus 0.68 for CatBoost).

3.5 Extensions (bonus)

An extension shows up as a second step: the below-standby hours become wider than the season explains.

One case misled the model. A small system already present at the start of the data lowers the withdrawal a little, but never below standby. Once extended, it brings the midday withdrawal to zero, and the model took the extension for the installation. Simulated cases of this kind were misdated 90% of the time, against 7% for ordinary extensions. A new rule checks the period before the detected start. If the midday withdrawal already follows the weather and drops with the shape of a sunny day, a small system was already there: the installation moves to the start of the data, and the detected start becomes an extension. It fires wrongly on 2 of 268 simulated systems and on no labelled one.

We report the rule's extensions plus the two strongest candidates of the detector in the whole file. The bonus is worth up to 10 points, $F_1^{\text{ext}} \times (7 + 1.5D^{\text{ext}} + 1.5P^{\text{ext}})$, and an extension counts as found when its date is within 30 days. The labelled data cannot measure it (their only extension comes 48 days before the end of the series), so we built simulated test files with the hidden test's composition given in the challenge (11 PV customers, one with one extension and one with two): the expected bonus is 2.7–3.0 points.

4 Using the unlabelled data

We first split the 1,147 unlabelled customers (two of the 1,149 series are not customers) into three groups, automatically and with no manual labelling. For each customer we computed the same score as the main rule (Section 3.2): the highest share of PV-like days in any 60-day window. We kept only those far from the 12.5% threshold, where the answer is clear:

- *clear PV* (213): a score of at least 37.5%, three times the threshold. No labelled non-PV customer exceeds 10%;
- *clear non-PV* (345): a score of at most 2%, a midday drop no larger than that of the median labelled customer without PV, and at least a year of data;
- *uncertain* (589): all the others, left aside.

The first two groups did most of the work:

- the clear-PV customers gave the weather index and, together with the labelled systems, the shape of a sunny day and the daily sunshine index;
- the clear non-PV customers hosted the simulated systems, including the hard cases: late installations, weak systems later extended, panels facing east or west, capped inverters;
- comparing the clear-PV customers day by day revealed two groups with different sunny days (their weather correlates only about 0.4). All labelled systems belong to one group, so this did not change our score. But 36 of the 196 clear-PV customers with enough data follow neither group's weather, which led to the weather check.

5 Results

Step (leave-one-out, 40 customers)	Score	F_1	D	P
Starting point: main rule, simple date, linear capacity	80.53	0.952	0.707	0.264
+ second rule for small systems	83.09	0.977	0.717	0.287
+ capacity model trained on simulated systems	85.86	0.977	0.717	0.476
+ shape check on the second rule	87.10	1.000	0.685	0.455
+ date fixes for weak starts	87.70	1.000	0.732	0.448
+ CatBoost, safety checks, rule for weak systems later extended	87.81	1.000	0.732	0.456
+ average of five capacity models, sunny-day drop as input	88.26	1.000	0.732	0.486
+ initial capacity from the vision transformer	**89.39**	**1.000**	**0.732**	**0.561**

Table 1: Leave-one-out score, step by step. The safety checks and the extension rule change no labelled prediction.

On the 40 labelled customers the final model finds all 22 systems and raises no false alarm. Capacity has a median error of 19%, with 16 of 22 within 25%. For comparison, our detection and dates with the same median capacity for everyone score 87.20: the capacity model adds about two points, and saturation limits the rest (Section 3.4).

How much can these numbers be trusted? Resampling the 40 customers gives a 95% interval of 83.0–93.9, and simulated 20-customer tests give 86.2–92.6. Both are optimistic, because we made choices while looking at the same 40 customers. We expect a lower score on the hidden test, plus 2.7–3.0 bonus points.

6 Challenges and what did not work

Main difficulties: few labels (only 18 customers without PV), addressed with simulated systems and unlabelled data; saturation, which hides the size of large systems; weak starts, which delayed the dates. These ideas failed our checks:

Idea	What we found
Capacity from the hour when withdrawal hits zero (production then equals the load)	Good ranking (correlation 0.86), but the scale is off by about ±40%; P fell from 0.455 to 0.388
Winter inputs (large systems do not saturate in winter); predicting kWp relative to the load	P fell from 0.448 to 0.393 and 0.396
More varied or more simulated systems	Better on simulated data, worse on real data (P down by 0.03–0.05)
External labelled data: HEAPO, Zurich homes (CC BY 4.0, arXiv:2503.16993)	67% false alarms, capacity twice too high: homes are 30–50 times smaller than AIL buildings
Stricter threshold near the end of the data, against spurious late detections	A threshold of 0.30 would drop one labelled late installation and 25% of simulated ones

7 AI agents and LLM usage

We used agentic coding tools to write code, brainstorm the approach and draft this report. The delivered pipeline contains no LLM.